Reporting Aggregated Data Using the Group Function

Assessment Questions---

01.

Using Single-Row_SQL_ass.sql | Welcome Page | samplesql

Worksheet | Query Builder

```sql
/*Question 01.
Find the highest, lowest, sum, and average salary of all employees. Label the columns

Query:-
*/
SELECT MAX(SAL) AS Maximum,
       MIN(SAL) AS Minimum,
       SUM(SAL) AS Sum,
       AVG(SAL) AS Average
FROM EMP;
```

Query Result

SQL | All Rows Fetched: 1 in 0.002 seconds

	MAXIMUM	MINIMUM	SUM	AVERAGE
1	5000	800	29025	2073.2142857142857142857142857142857143

02.

```sql
/*Question 02.
Maximum, Minimum, Sum, and Average, respectively. Round your results to the nearest whole number.

Query:-
*/
SELECT ROUND(MAX(SAL)) AS Maximum,
       ROUND(MIN(SAL)) AS Minimum,
       ROUND(SUM(SAL)) AS Sum,
       ROUND(AVG(SAL)) AS Average
FROM EMP;
```

Query Result

SQL | All Rows Fetched: 1 in 0.002 seconds

	MAXIMUM	MINIMUM	SUM	AVERAGE
1	5000	800	29025	2073

03.

```sql
/*Question 03.
Modify the above query to display the minimum, maximum, sum, and average salary for each job type.

Query:-
*/
SELECT JOB,
       ROUND(MIN(SAL)) AS Minimum,
       ROUND(MAX(SAL)) AS Maximum,
       ROUND(SUM(SAL)) AS Sum,
       ROUND(AVG(SAL)) AS Average
FROM EMP
GROUP BY JOB;
```

Query Result

All Rows Fetched: 5 in 0.004 seconds

	JOB	MINIMUM	MAXIMUM	SUM	AVERAGE
1	CLERK	800	1300	4150	1038
2	SALESMAN	1250	1600	5600	1400
3	PRESIDENT	5000	5000	5000	5000
4	MANAGER	2450	2975	8275	2758
5	ANALYST	3000	3000	6000	3000

04.

```sql
/*Question 04.
Write a query to display the number of people with the same job

Query:-
*/
SELECT JOB,
       COUNT(*) AS Number_of_Employees
FROM EMP
GROUP BY JOB;
```

Query Result

All Rows Fetched: 5 in 0.002 seconds

	JOB	NUMBER_OF_EMPLOYEES
1	CLERK	4
2	SALESMAN	4
3	PRESIDENT	1
4	MANAGER	3
5	ANALYST	2

05.

```sql
/*Question 05.
Determine the number of managers without listing them. Label the column Number of Managers

Query:-
*/
SELECT COUNT(DISTINCT MGR) AS "Number of Managers"
FROM EMP;
```

Query Result ×

SQL | All Rows Fetched: 1 in 0.006 seconds

	Number of Managers
1	6

06.

```sql
/*Question 06.
Find the difference between the highest and lowest salaries. Label the column DIFFERENCE.

Query:-
*/
SELECT MAX(SAL) - MIN(SAL) AS DIFFERENCE
FROM EMP;
```

Query Result ×

SQL | All Rows Fetched: 1 in 0.003 seconds

	DIFFERENCE
1	4200

07.

```sql
/*Question 07.
Create a report to display the manager number and the salary of the lowest-paid employee for that manager. Exclude anyone

Query:-
*/
SELECT MGR,
       MIN(SAL) AS Lowest_Salary
FROM EMP
WHERE MGR IS NOT NULL
GROUP BY MGR
HAVING MIN(SAL) > 2000
ORDER BY Lowest_Salary DESC;
```

Query Result ×

SQL | All Rows Fetched: 2 in 0.001 seconds

	MGR	LOWEST_SALARY
1	7566	3000
2	7839	2450

08.

```sql
/*Question 08.
Create a query to display the total number of employees and, of that total, the number of employees hired in 1980, 1981, and 1982. Create

Query:-
*/
SELECT COUNT(*) AS Total_Employees,
       SUM(CASE WHEN TO_CHAR(HIREDATE,'YYYY')='1980' THEN 1 ELSE 0 END) AS "Hired in 1980",
       SUM(CASE WHEN TO_CHAR(HIREDATE,'YYYY')='1981' THEN 1 ELSE 0 END) AS "Hired in 1981",
       SUM(CASE WHEN TO_CHAR(HIREDATE,'YYYY')='1982' THEN 1 ELSE 0 END) AS "Hired in 1982"
FROM EMP;
```

Query Result ×

SQL | All Rows Fetched: 1 in 0.001 seconds

	TOTAL_EMPLOYEES	Hired in 1980	Hired in 1981	Hired in 1982
1	14	1	10	1